Adding and Subtracting Unlike Fractions
Answer Key
Grade 5 Math

Quick Answers

1. 4
2. 9
3. $\frac{5}{6}$, $\frac{3}{8}$
4. $\frac{3}{4}$
5. $\frac{5}{6}$
6. $\frac{7}{12}$
7. $\frac{3}{8}$, $\frac{5}{12}$
8. $\frac{23}{24}$
9. $\frac{8}{15}$
10. $\frac{23}{6}$
11. $\frac{35}{12}$
12. $\frac{17}{12}$

Curriculum

Level: Grade 5 Math
4.NF.A – *problems 1, 2, 3, 7*
5.NF.A.1 – *problems 4, 5, 6, 8, 9, 10, 11, 12*
Difficulty 1–4 of 5 across 14 tagged checks.

1. Rewrite $\frac{1}{2}$ as an equivalent fraction with a denominator of 8. What is the missing numerator?

Solution: $8 \div 2 = 4$, so the denominator was multiplied by 4. Multiply the numerator by the same factor: 1×4.

$\Rightarrow$ $\boxed{n = 4}$

2. Fill in the missing numerator: $\frac{3}{4} = \frac{?}{12}$

Solution: $12 \div 4 = 3$, so multiply the numerator by 3 as well: 3×3.

$\Rightarrow$ $\boxed{n = 9}$

3. Rewrite $\frac{5}{6}$ and $\frac{3}{8}$ as equivalent fractions with the common denominator 24.

Solution: $24 \div 6 = 4$, so multiply the top and bottom of $\frac{5}{6}$ by 4.

$\Rightarrow$ $\boxed{\frac{20}{24}}$

Then $24 \div 8 = 3$, so multiply the top and bottom of $\frac{3}{8}$ by 3.

$\Rightarrow$ $\boxed{\frac{9}{24}}$

4. Add. $\dfrac{1}{2}+\dfrac{1}{4}$

Solution: 4 is already a multiple of 2, so the common denominator is 4: $\dfrac{1}{2}=\dfrac{2}{4}$. Then $\dfrac{2}{4}+\dfrac{1}{4}$ adds the numerators over the common denominator.

$\Rightarrow$ $\boxed{\frac{3}{4}}$

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5. Add. $\dfrac{2}{3}+\dfrac{1}{6}$

Solution: 6 is a multiple of 3, so use 6: $\dfrac{2}{3}=\dfrac{4}{6}$. Then $\dfrac{4}{6}+\dfrac{1}{6}=\dfrac{5}{6}$, which is already in lowest terms.

$\Rightarrow$ $\boxed{\frac{5}{6}}$

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6. Subtract. $\dfrac{3}{4}-\dfrac{1}{6}$

Solution: The least common denominator of 4 and 6 is 12: $\dfrac{3}{4}=\dfrac{9}{12}$ and $\dfrac{1}{6}=\dfrac{2}{12}$. Subtract the numerators: $9-2=7$.

$\Rightarrow$ $\boxed{\frac{7}{12}}$

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7. Find the least common denominator of $\dfrac{3}{8}$ and $\dfrac{5}{12}$, then rewrite each fraction with it.

Solution: Multiples of 8: $8, 16, 24, \ldots$ Multiples of 12: $12, 24, \ldots$ The least common denominator is 24. Since $24 \div 8 = 3$, multiply $\dfrac{3}{8}$ top and bottom by 3.

$\Rightarrow$ $\boxed{\frac{9}{24}}$

Since $24 \div 12 = 2$, multiply $\dfrac{5}{12}$ top and bottom by 2.

$\Rightarrow$ $\boxed{\frac{10}{24}}$

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8. Add. $\dfrac{5}{8}+\dfrac{1}{3}$

Solution: 8 and 3 share no factor, so the least common denominator is $8\times 3=24$: $\dfrac{5}{8}=\dfrac{15}{24}$ and $\dfrac{1}{3}=\dfrac{8}{24}$. Add the numerators: $15+8=23$.

$\Rightarrow$ $\boxed{\frac{23}{24}}$

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9. Subtract. $\dfrac{5}{6}-\dfrac{3}{10}$

Solution: The least common denominator of 6 and 10 is 30: $\dfrac{5}{6}=\dfrac{25}{30}$ and $\dfrac{3}{10}=\dfrac{9}{30}$. Subtract: $25-9=16$, giving $\dfrac{16}{30}$. Both parts divide by 2.

$\Rightarrow$ $\boxed{\frac{8}{15}}$

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10. Add the mixed numbers. $2\,\frac{1}{2} + 1\,\frac{1}{3}$

Solution: Add the whole numbers: $2 + 1 = 3$. For the fractions use the common denominator 6: $\frac{1}{2} = \frac{3}{6}$ and $\frac{1}{3} = \frac{2}{6}$, so the fraction part is $\frac{5}{6}$.

$\Rightarrow$ $\boxed{3\,\frac{5}{6}}$

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11. Subtract the mixed numbers. $4\,\frac{3}{4} - 1\,\frac{5}{6}$

Solution: Common denominator 12: $4\,\frac{3}{4} = 4\,\frac{9}{12}$ and $1\,\frac{5}{6} = 1\,\frac{10}{12}$. Since $\frac{9}{12}$ is smaller than $\frac{10}{12}$, regroup one whole: $4\,\frac{9}{12} = 3\,\frac{21}{12}$. Now subtract whole from whole and fraction from fraction: $3 - 1 = 2$ and $\frac{21}{12} - \frac{10}{12} = \frac{11}{12}$.

$\Rightarrow$ $\boxed{2\,\frac{11}{12}}$

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12. A muffin recipe uses $\frac{2}{3}$ of a cup of flour and $\frac{3}{4}$ of a cup of oats. How many cups of dry ingredients in all?

Solution: Total means add: $\frac{2}{3} + \frac{3}{4}$. The least common denominator of 3 and 4 is 12: $\frac{2}{3} = \frac{8}{12}$ and $\frac{3}{4} = \frac{9}{12}$. Adding gives $\frac{17}{12}$, which is more than one whole, so rename it as a mixed number: $\frac{17}{12} = 1\,\frac{5}{12}$ cups.

$\Rightarrow$ $\boxed{1\,\frac{5}{12}}$